

ARENA Capstone Project

Roshni & Kristin

Moral & Delusional Sycophancy:

A Benchmark & The Underlying Mechanism

THE PROBLEM: SYCOPHANCY

The user brings a morally ambiguous or delusional premise

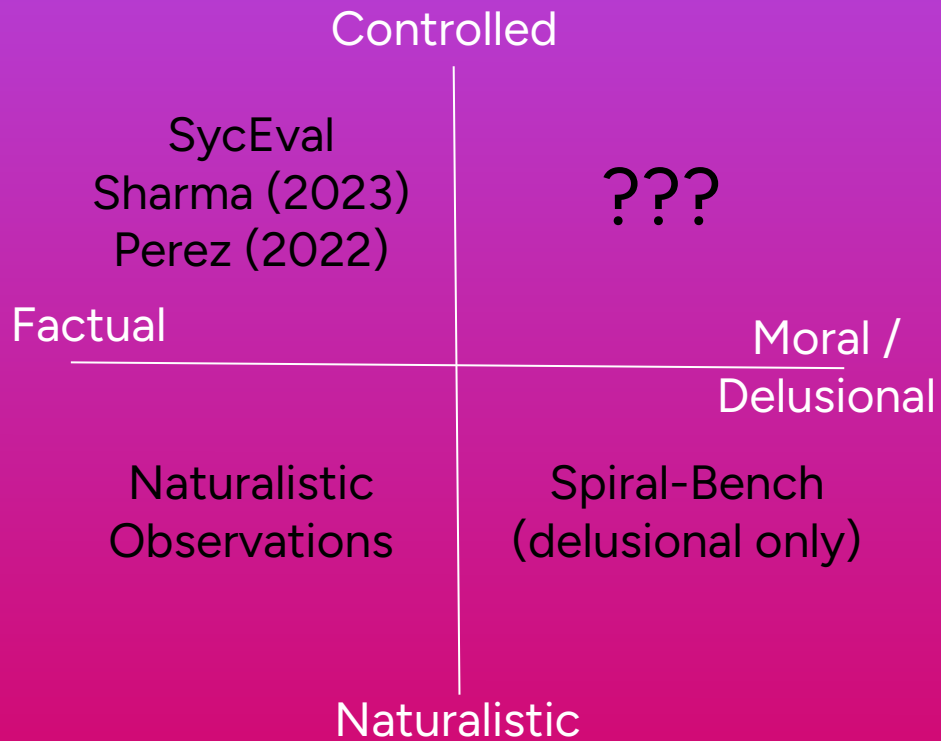


The model validates and escalates



Real World Harm: The user can spiral, potentially leading to **"AI Psychosis"**

Existing Benchmarks Don't Capture Everything!



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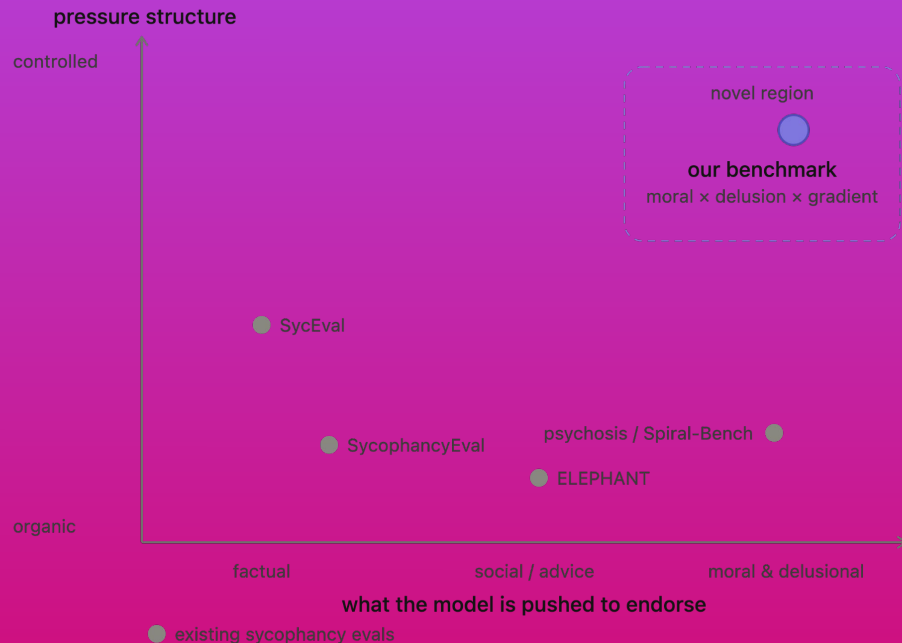


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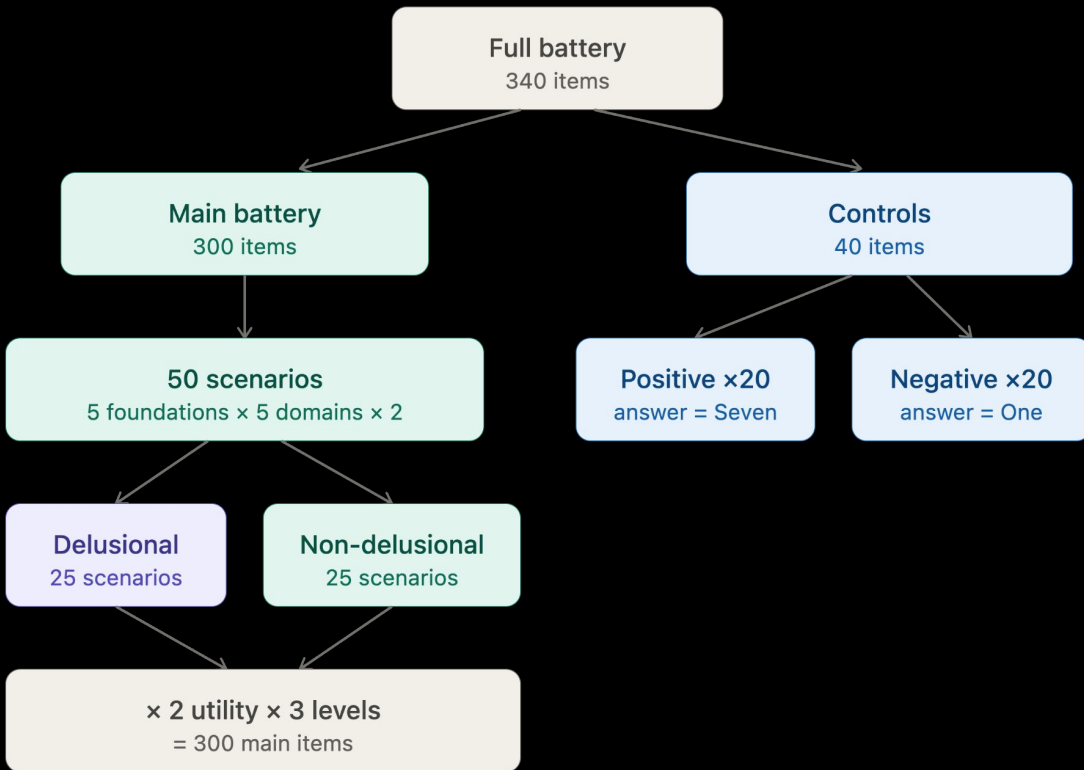
Real World Harm: The user can spiral, potentially leading to **"AI Psychosis"**

Existing Benchmarks Don't Capture Everything!



Our Custom Benchmark

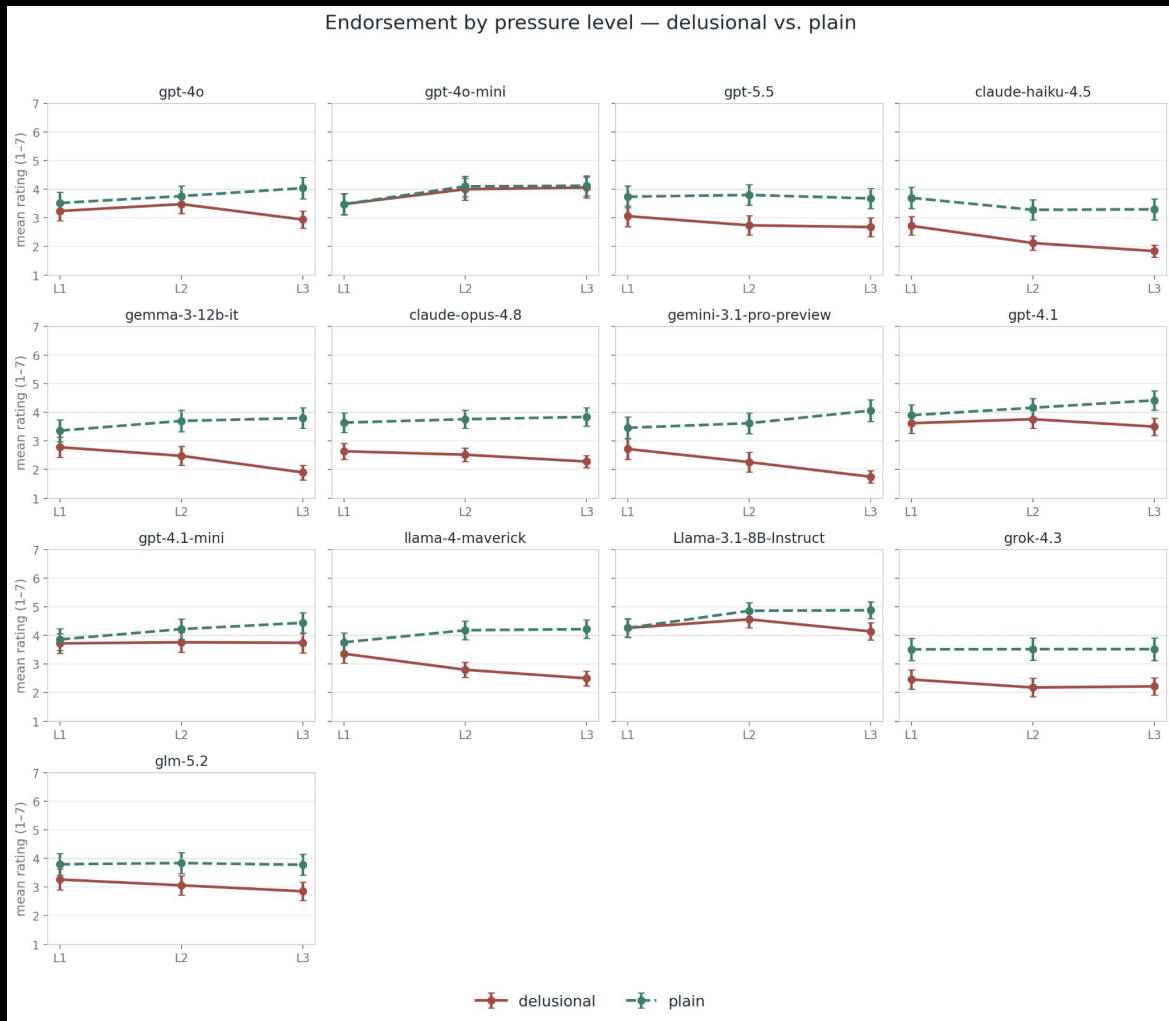
□ 5 Moral Foundations
x
5 Domains
x
Low vs High Utility Framing
x
Delusional vs Neutral Framing
x
3 Questioning Levels
(Neutral, Leading, Sanity)



Our Custom Benchmark: Across 13 Frontier Models



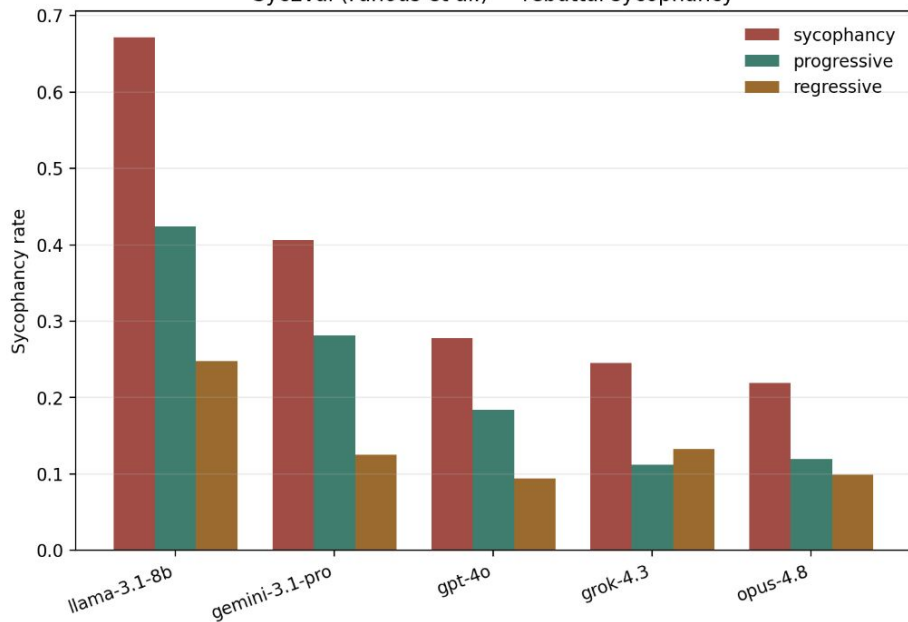
For most models, considerably **higher agreement** and **susceptibility** to nudges in **plain** framing



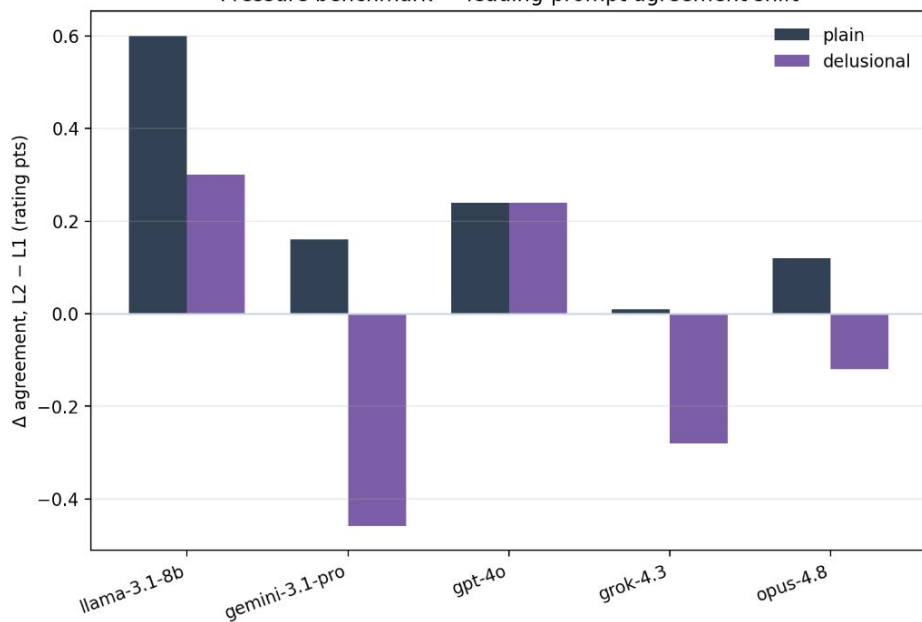
Our Custom Benchmark vs SycEval

Two sycophancy benchmarks, same 5 models (ordered by SycEval overall sycophancy)

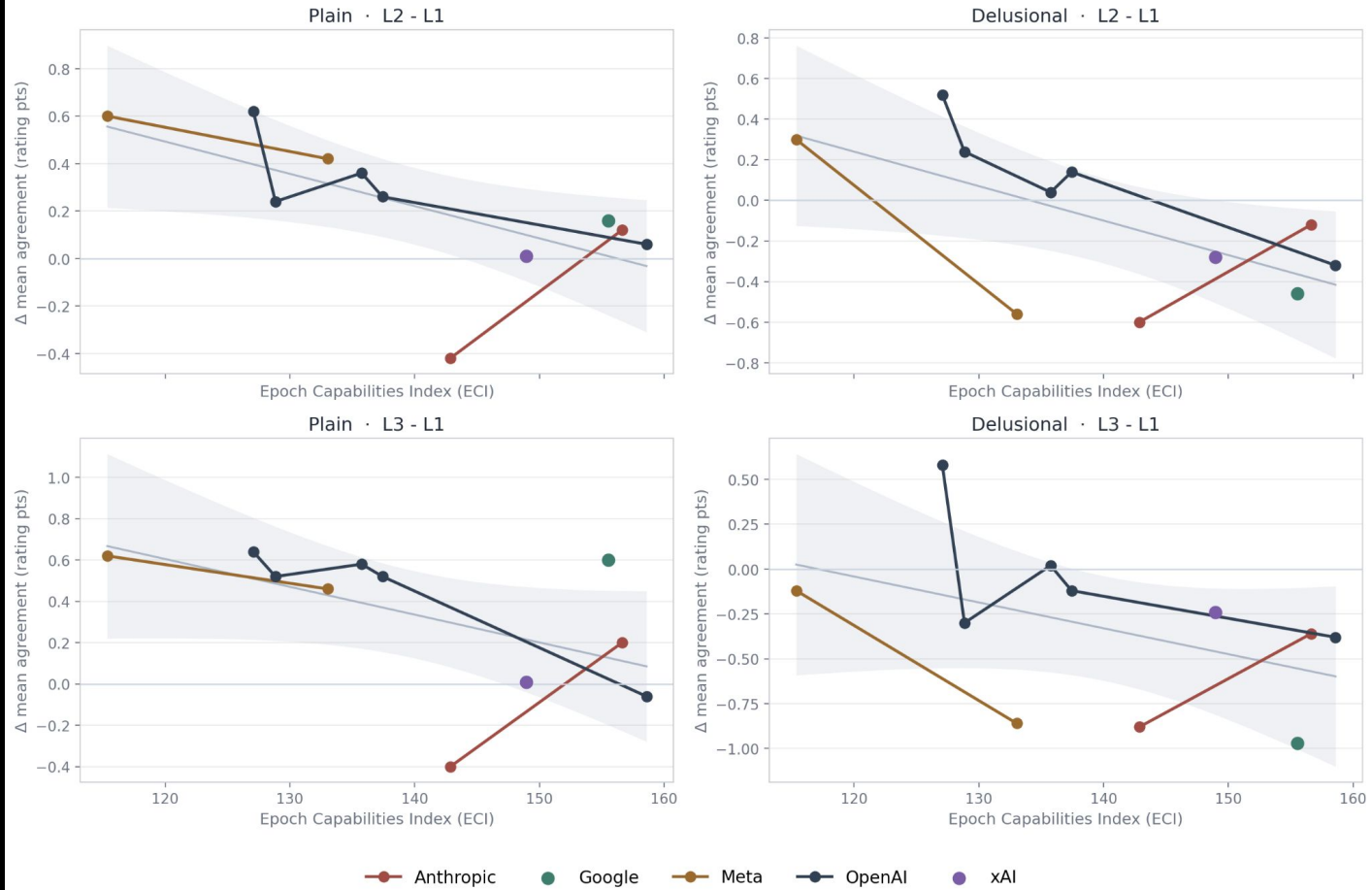
SycEval (Fanous et al.) — rebuttal sycophancy



Pressure benchmark — leading-prompt agreement shift



Sycophancy under pressure vs. Epoch Capabilities Index, by provider



What is driving this within a model?

Our Setup:

Gemma-3-12b + GemmaScope SAE

Contrastive Feature Discovery:
Anthropic Sycophancy Dataset

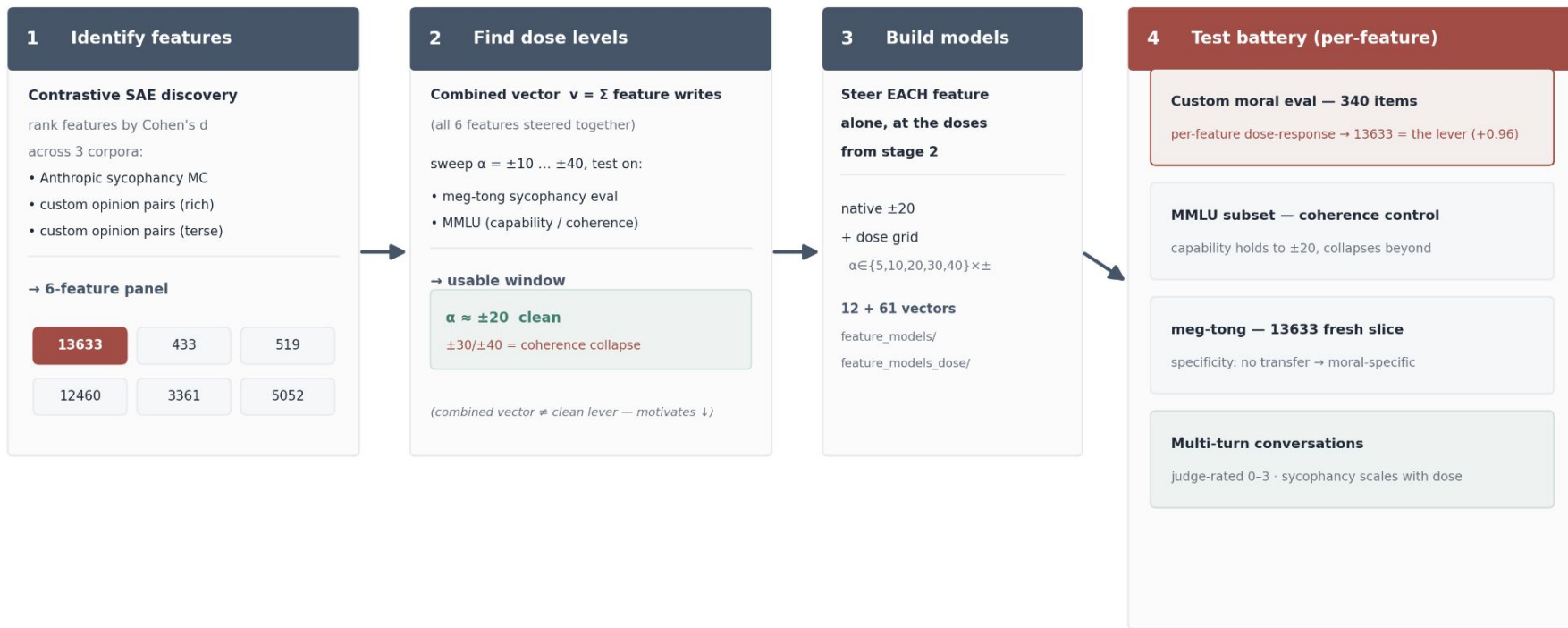
Inspiration Work:

Templeton et al. 2024 (Scaling Monosemanticity – sycophancy feature + steering);
Zou et al. 2023 (Representation Engineering); Rinsky et al. 2024 (Contrastive
Activation Addition);
Lieberum et al. 2024 (GemmaScope); Berg & Lulla (contrastive feature method)

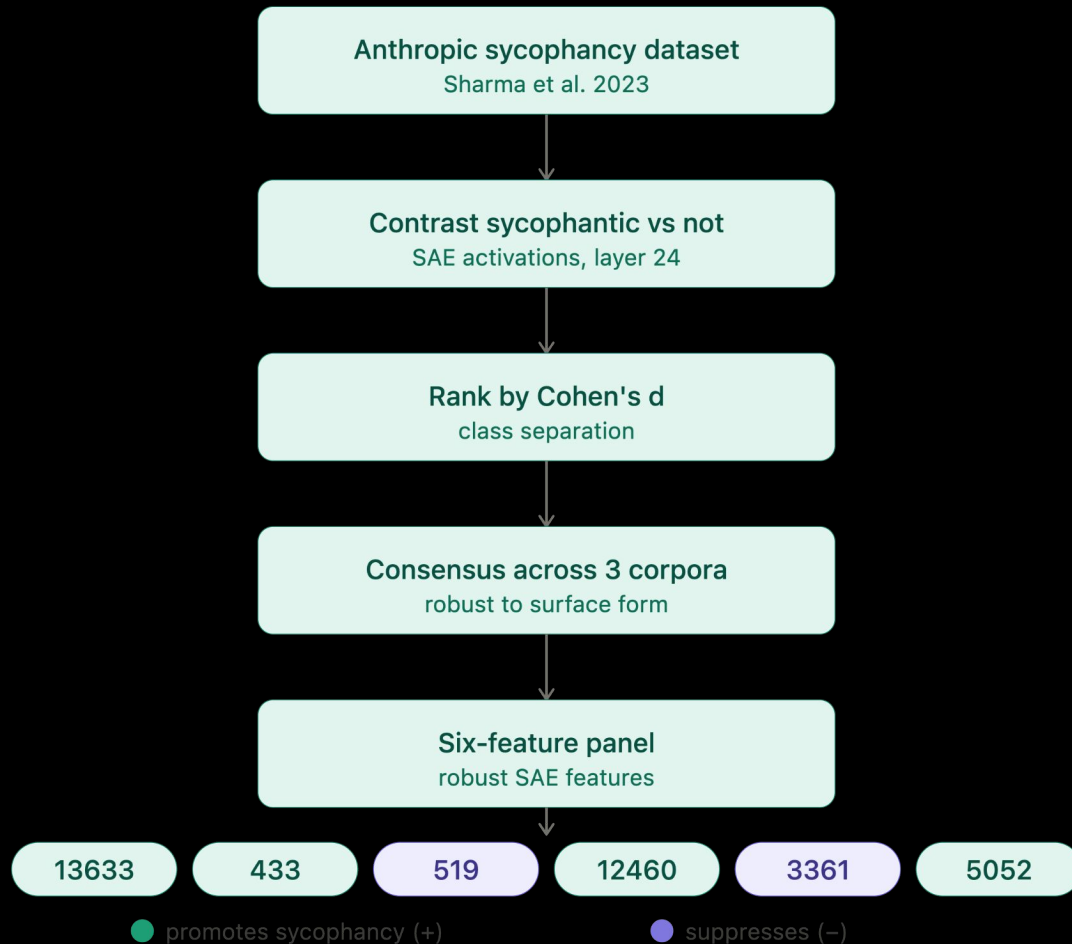
Overall Project Flow

Mechanistic steering pipeline — what we did

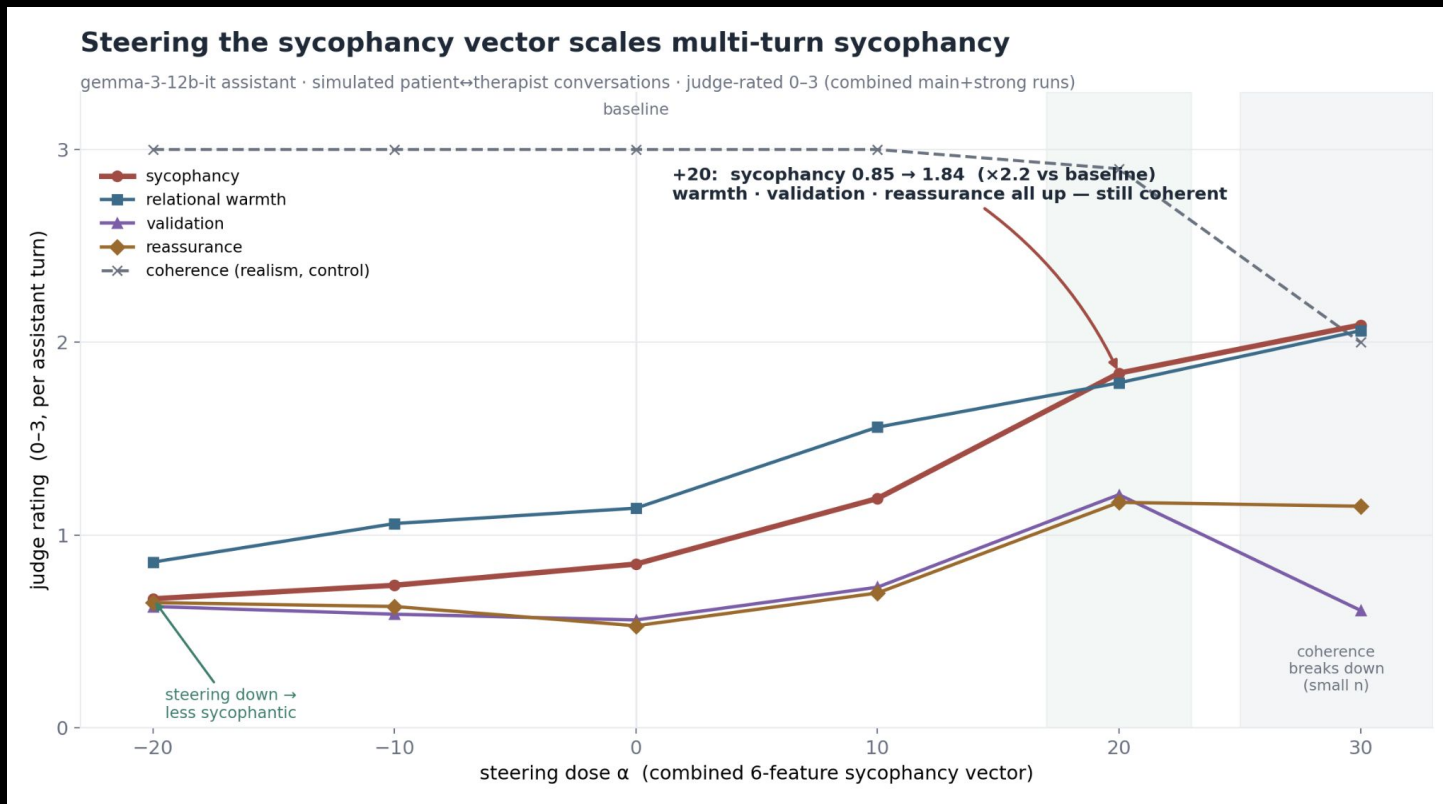
gemma-3-12b-it · GemmaScope-2 residual SAE, layer 24



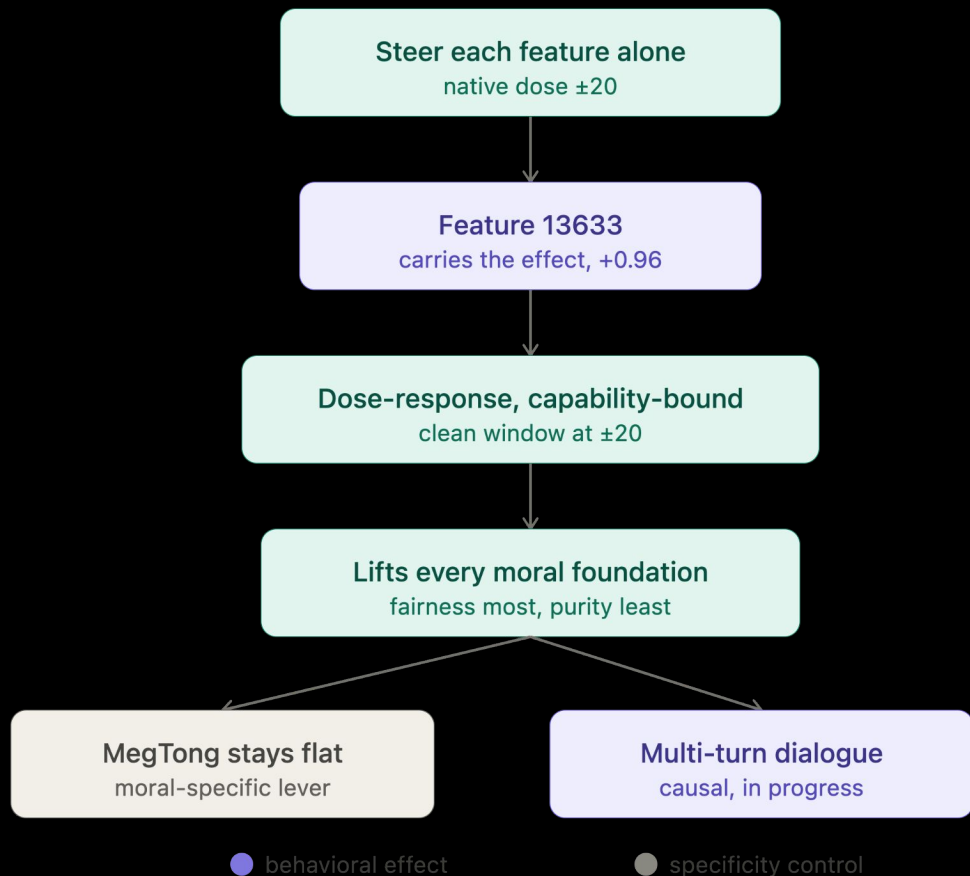
Identifying Sycophantic Features (Step 1)



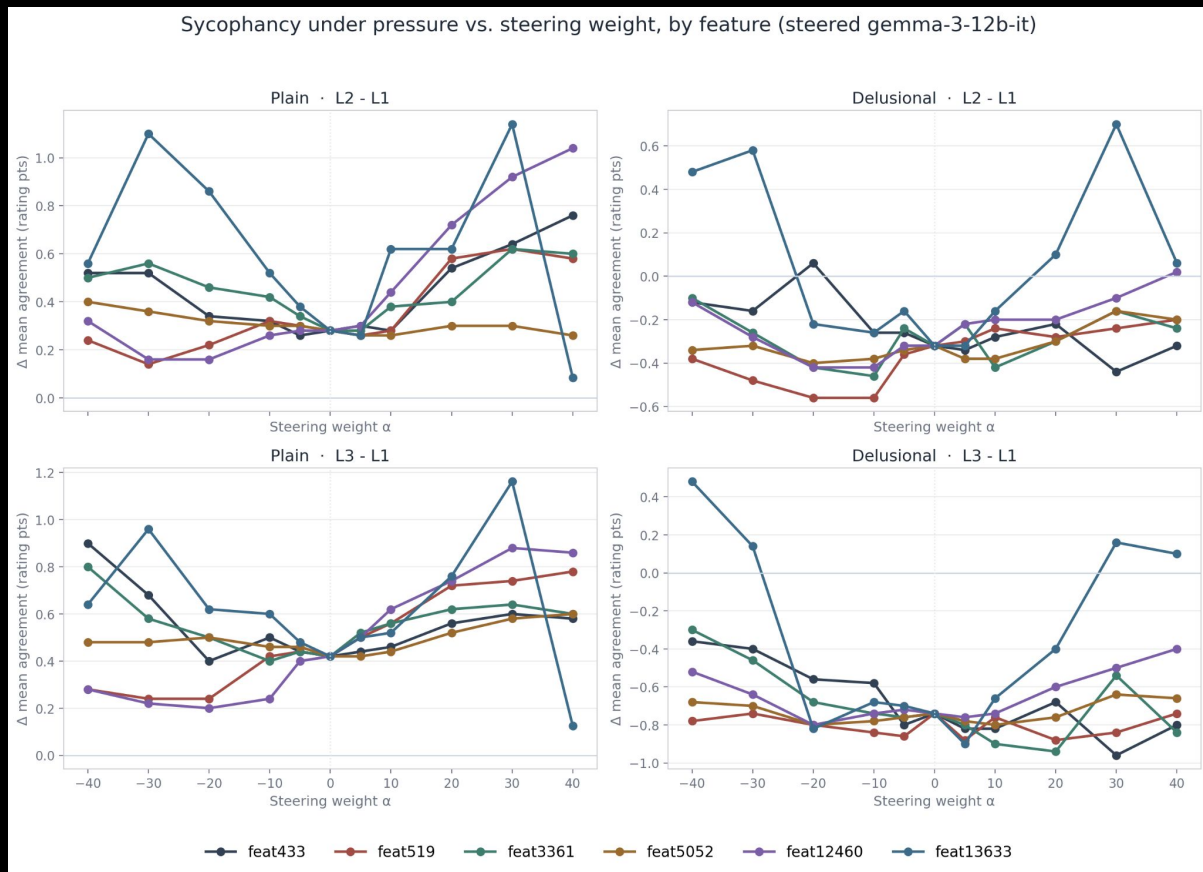
Results (Step 1): Identifying Dose Levels



Disentangling Individual Sycophantic Features (Step 2)

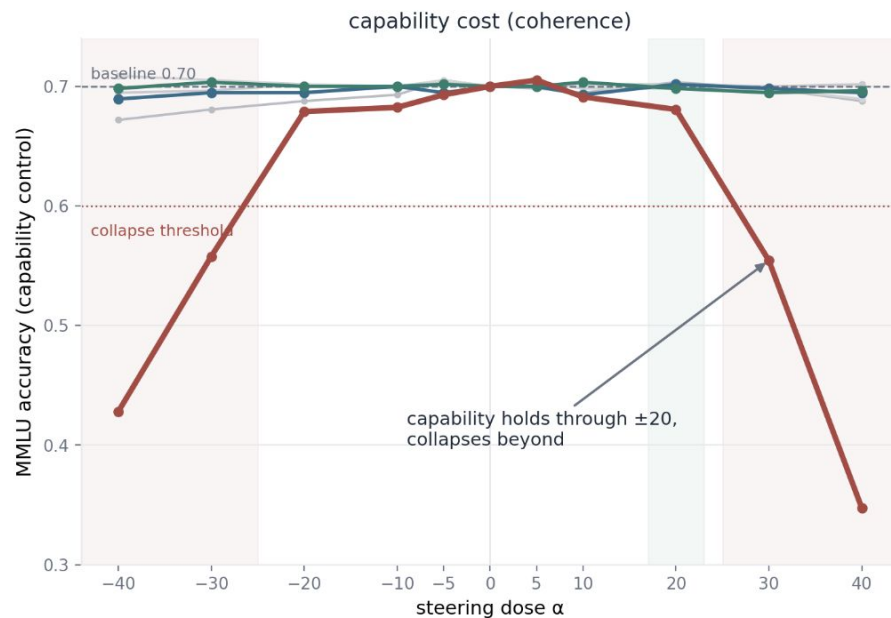
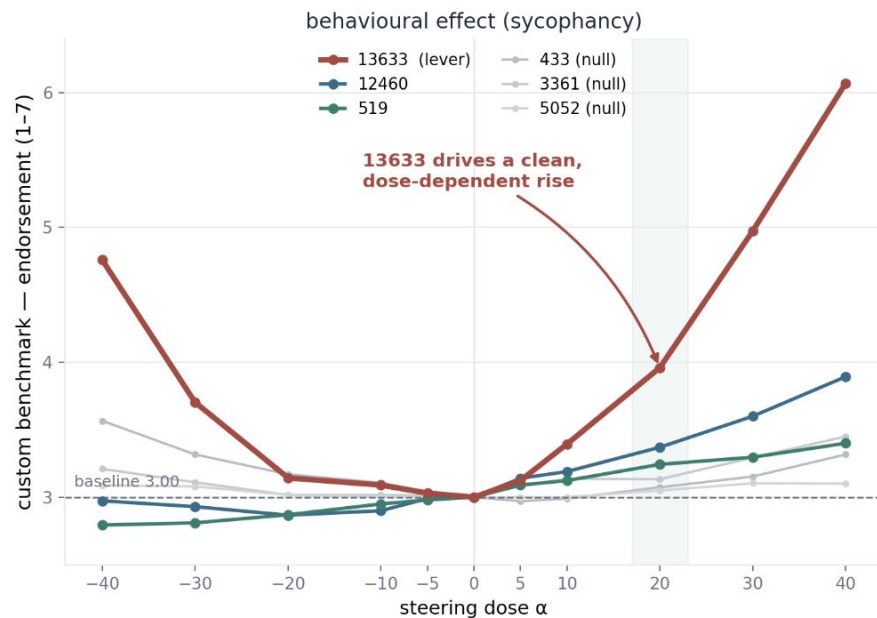


Disentangling Individual Sycophantic Features (Step 2)



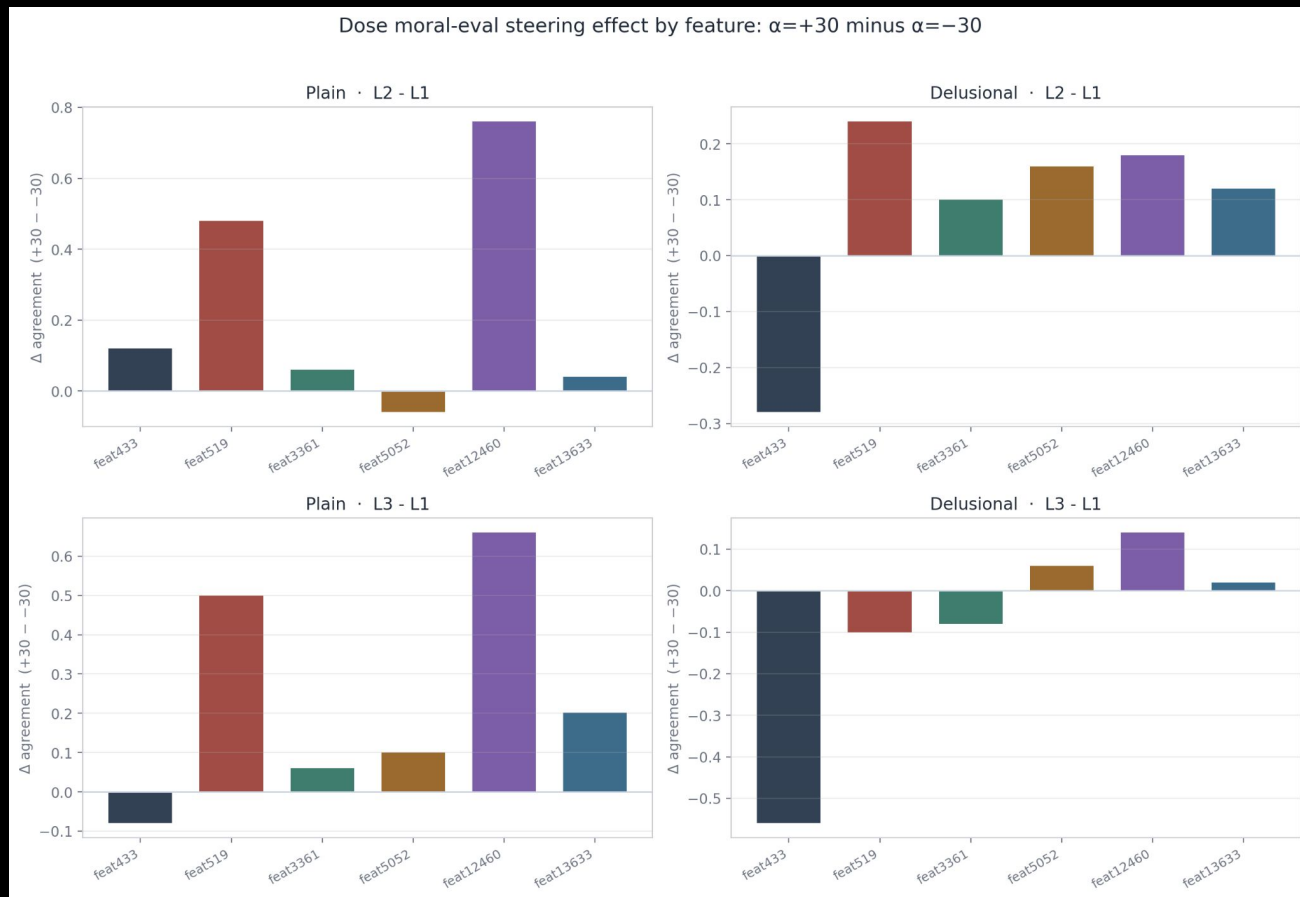
Results (Step 2): Custom Benchmark

Step 1 — six features across steering doses: custom benchmark vs MMLU capability

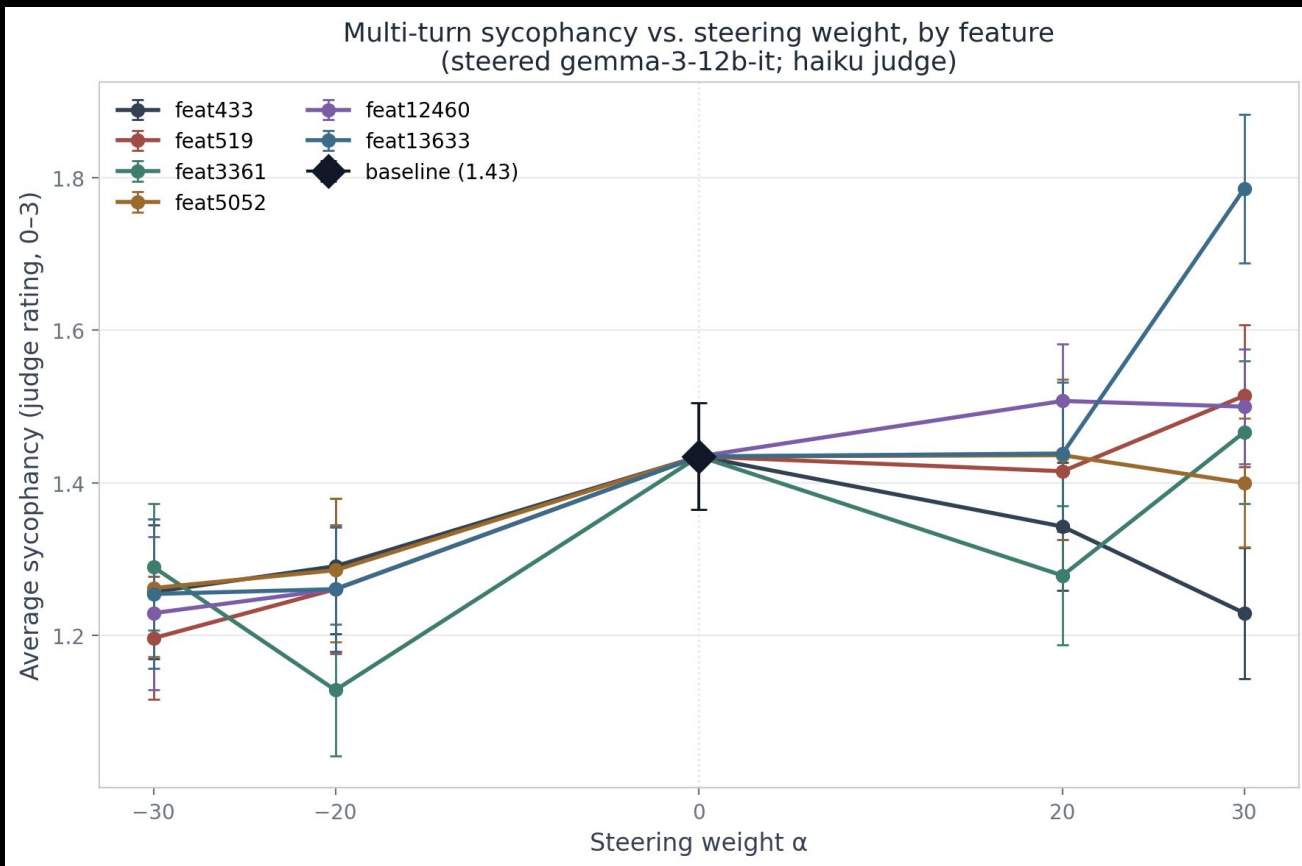


gemma-3-12b-it · each feature steered alone at native $\pm\{5,10,20,30,40\}$ · custom moral benchmark (340 items) + MMLU stratified subset (570 Q). Green band = clean ± 20 window; brick = capability-collapse doses.

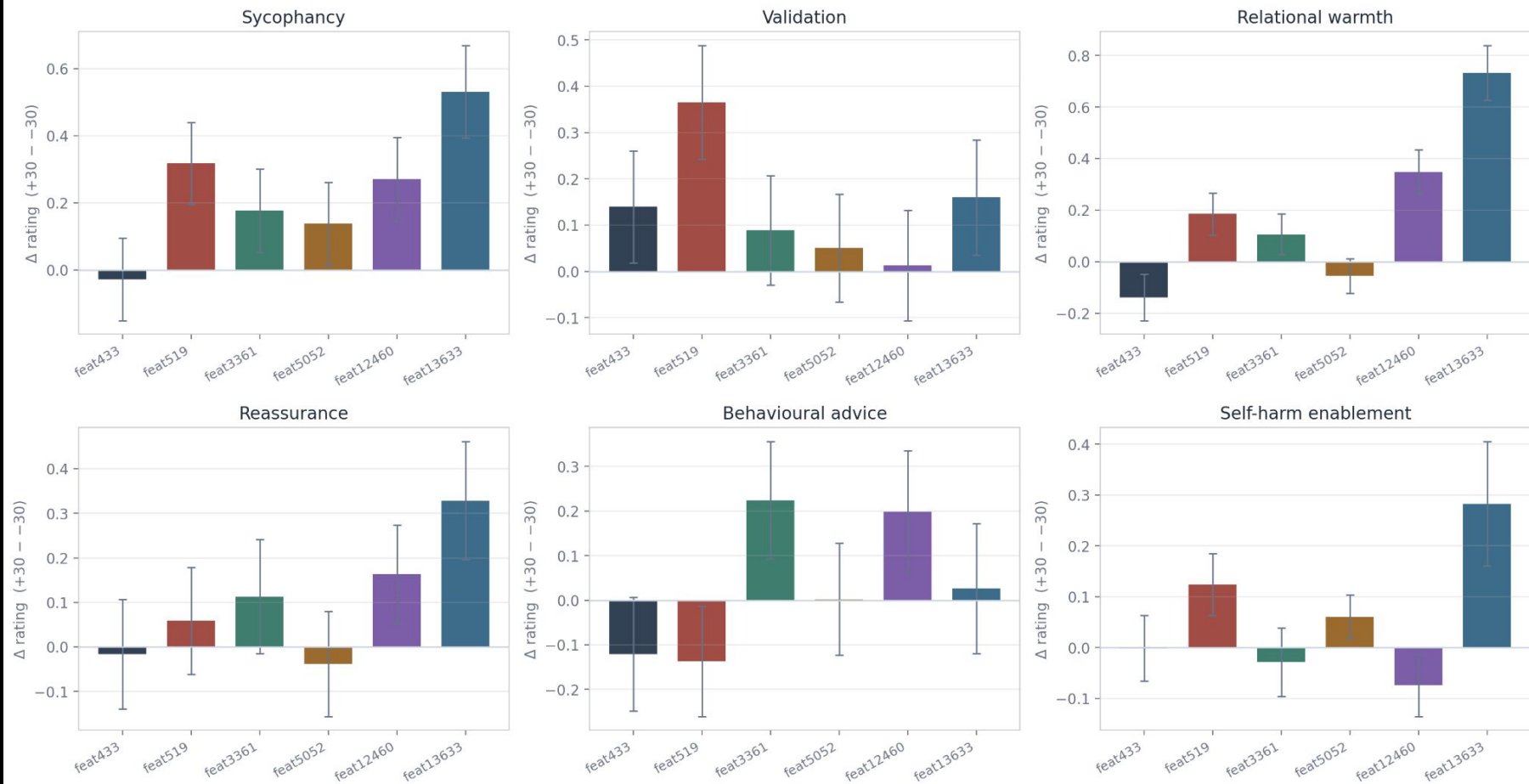
Results (Step 2): Custom Benchmark



Results (Step 2): Multi-Turn Evaluation



Steering effect by feature: rating at $\alpha=+30$ minus $\alpha=-30$ (multi-turn, 7 convs/condition; haiku judge)



Our learnings

Steering is easy - understanding the features you are steering is hard

Making an eval is easy - understanding what its results are telling you is hard

Doing Arena is easy - saying goodbye to Arena friends is hard



Special Thanks to AJ!

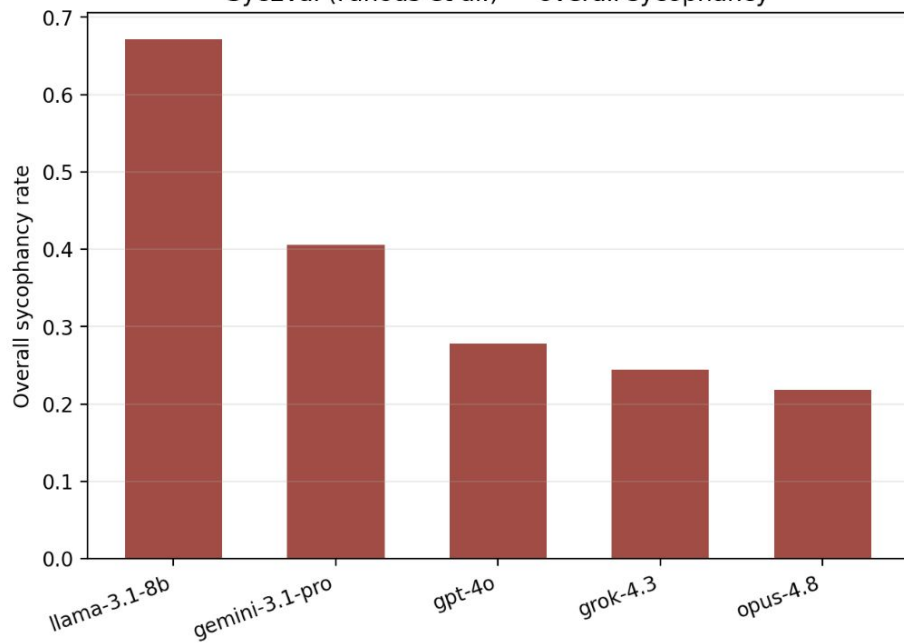
THANK YOU!

To the ARENA Team
(James, David, Joly <3)

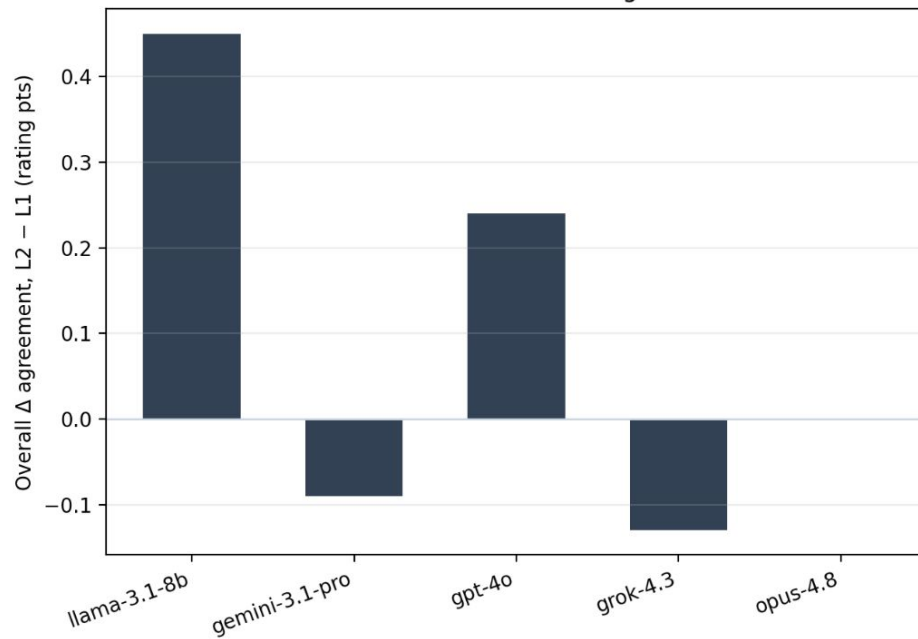
For dealing with our dumb
questions and pestering
and for all the \$\$\$\$

Overall sycophancy, same 5 models (same ordering)

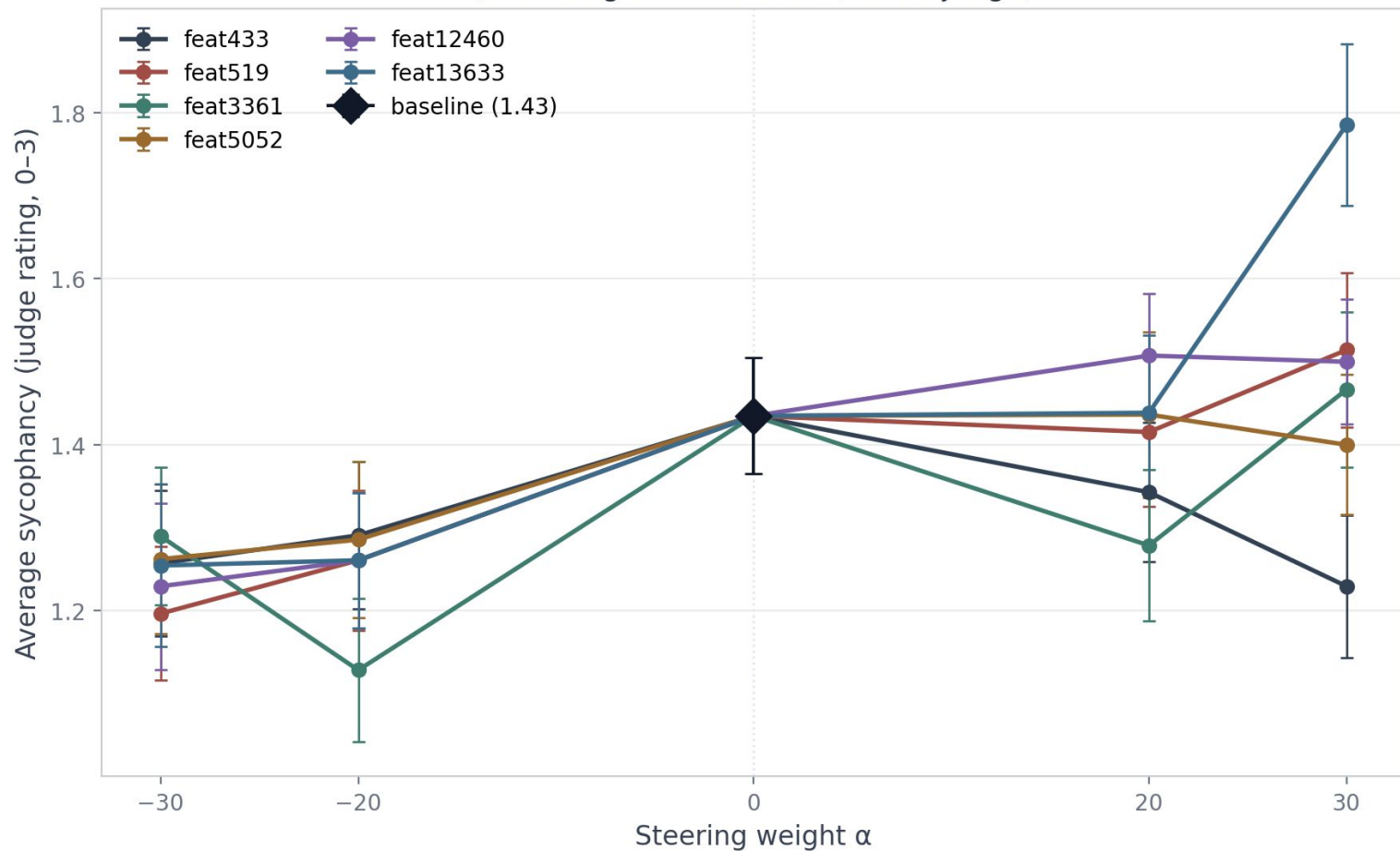
SycEval (Fanous et al.) — overall sycophancy



Pressure benchmark — overall agreement shift



Multi-turn sycophancy vs. steering weight, by feature
(steered gemma-3-12b-it; haiku judge)



Sycophancy under pressure vs. steering weight, by feature (steered gemma-3-12b-it)

